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Effect of hatching distance and scanning path on $MgAl_2O_4$/Ti6Al4V joint by femtosecond laser welding: Microstructure, mechanical properties and residual stress

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Abstract

The effect of hatching distance and scanning path on the microstructure and mechanical properties of $MgAl_2O_4$/Ti6Al4V joint has been elucidated. In our research, three kinds of scanning paths (serpentine, fillet-serpentine and concentric circle) and five hatching distances are adopted. Direct bonding of the concentric circle welding seam with the hatching distance of 0.025 mm by femtosecond laser showed the highest shear strength, which can reach 27.56 MPa. Nondestructive residual stress measurement by Raman spectra has been taken in this study, and the joint with concentric circle welding line possessed the minimum stress value of 22.83 MPa, which is consistent with the trend of the shear strength.

Keywords

Femtosecond laser; $MgAl_2O_4$ ceramic; Ti6Al4V alloy; Scanning path; Hatching distance

1. Introduction

With the unceasing development of the times, medical technology is also advancing by leaps and bounds. Endoscopes, as one of the most commonly used medical scope could not only be limited to diagnosis and examination, but also can be applied to treatment and surgery [[1], [2], [3]]. Ti6Al4V alloy, which is benefit from its outstanding mechanical properties and bio-compatibility [4], is usually chosen to manufacture various component parts in endoscopic equipment, such as shells, pipes, pincers and clips. $MgAl_2O_4$ ceramic is widely used as optical window materials such as detector window and prism because of its high strength, high hardness and conspicuous light transmission performance [5,6]. Therefore, these two are the best choices for fabricating endoscopes. The joining between ceramics and metals is an important technology for engineering [[7], [8], [9]], however, it is a thorny problem for realizing effective bonding of $MgAl_2O_4$ ceramic and Ti6Al4V alloy due to their great differences in physical and chemical properties.

The commonly used methods for joining ceramic and metal usually includes adhesive bonding [10], rivet bonding [11], brazing [12] and diffusion bonding [13]. Notwithstanding adhesive bonding has the advantage of low cost and simple operation, the strength of the joint is relatively low which makes it difficult to serve in harsh environment. The brittleness of ceramics causes riveting not the best joining method. Likewise, brazing and diffusion bonding usually need superior requirements for environment, and both of them are time-consuming, which goes ill with efficient production. In recent years, femtosecond laser shows its unique charm in transparent material welding thanks to its ultra-fast and super-strong characteristics [14]. So-called ultrafast, it means that its pulse width is only in the order of femtosecond (10^{-15} s), which is far less than the time for electrons to relax to the lattice in the material. Therefore, the generated thermal effect and thermal diffusion are far less than those of traditional laser welding. Super-strong trait refers to its ultra-high peak density, which can trigger the nonlinear absorption of transparent materials [15]. These two features make femtosecond laser welding possess great potential in ceramic-metal bonding.

In 2005, Tamaki et al. [16] applied femtosecond laser to the welding of transparent materials for the first time. They successfully joined two pieces of silica glass and verified that femtosecond laser welding can realize spatially localized welding because of the limitation of thermal effect around the focal region, which greatly reduced the damage to the materials. Thereafter, femtosecond laser was gradually used in transparent materials welding, but the base material is also mostly glass [[17], [18], [19]]. Until 2022, Lipateva et al. [20] used femtosecond laser to weld Nd:YAG transparent ceramics, which proved that femtosecond laser can also weld transparent crystal materials with higher melting point, and introduced it into the field of transparent ceramic welding. In 2023, Pan et al. [21] successfully achieved the effective bonding between sapphire and Fe-36Ni alloy with the high shear strength, which can reach up to 108.35 MPa. This undoubtedly proves that femtosecond laser welding has great potential in realizing the joining between transparent ceramics and metals, and even can obtain a joint with higher strength and higher performance than traditional welding methods. By this token, femtosecond laser welding has great feasibility to join $MgAl_2O_4$ ceramic and Ti6Al4V alloy.

It is worth noting that, as a consensus, $MgAl_2O_4$ ceramic and Ti6Al4V alloy have discrepancies in thermal expansion coefficients resulting in residual stress in the joint, which is easy to damage the mechanical properties of the joint. Therefore, it is indispensable to optimize the welding mode to improve joint performance. A lot of research work has been carried out to investigate the effect of laser power [22], scanning speed [23], repetition rate [24], laser wavelength [25] on femtosecond laser welding. However, far too less attention has been paid to effect of scanning path and hatching distance on $MgAl_2O_4$/Ti6Al4V joint achieved by femtosecond laser welding, which also play an important role in the performance of the joint.

In this study, we successfully used femtosecond laser welding to join $MgAl_2O_4$ ceramic and Ti6Al4V alloy. The typical structure of the joint has been characterized and illustrated in detail. Additionally, the effect of hatching distance on the microstructure and shear strength were investigated thoroughly. Besides, three different scanning paths were adopted to achieve effective $MgAl_2O_4$/Ti6Al4V joint in parallel with the clarification of distinct scanning path effect on the mechanical properties of the joint. Afterwards, the residual stress of the joint was measured by Raman spectra, which aims to shed light on the failure behavior of the joint.

2. Experimental

Double-sided polished transparent $MgAl_2O_4$ with a size of 5 mm×5 mm×3 mm, was purchased from Hefei Crystal Technical Material Co., Ltd. The Ti6Al4V alloy was provided by Baoji Chuangxin Metal Materials Co., Ltd., and cut it into 12 mm×7 mm×3 mm pieces using an electric spark cutting machine. Afterwards, the surfaces to be welded of Ti6Al4V were ground step by step with different grades of sandpapers and then polished to ~1 μm finish. Before the welding procedure, the two base materials were put in ethanol and ultrasonically cleaned for 10 min.

Then the samples were assembled and tightened as shown in Fig. 1b until Newton's rings could be observed, the aforementioned stripes of light and dark, means that the gap between the two base materials is less than λ/4, which can restrict plasma and melt splash to guarantee the successful bonding. It is worth mentioning that the universal wheel equipped on the fixture can effectively prevent the joint from failing due to excessive torque during unloading after welding. The femtosecond laser can emit a minimum of 32 fs pulses with 800 nm, 1 kHz repetition rate and a maximum of 2.5 W average power. The whole assembly was then positioned on the five-axis process platform shown in Fig. 1a to start the welding process.

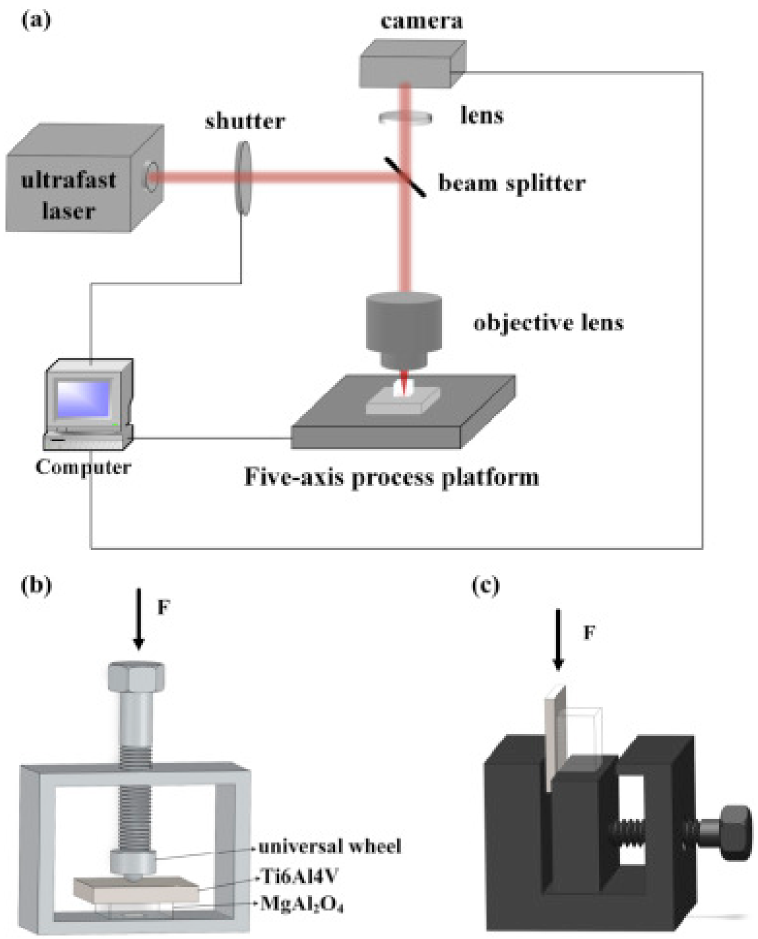

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Fig. 1. (a) Schematic diagrams of femtosecond laser welding, (b) the assembly of $MgAl_2O_4$ and Ti6Al4V samples, (c) shear strength test.

The piezoelectric coefficient of $MgAl_2O_4$ has not been reported yet, so this value needs to be measured. An PMMA resin plate with a size of 15 mm*8 mm*2.5 mm and a $MgAl_2O_4$ slice with a size of 6 mm*3 mm*0.1 mm are prepared and stacked up on the fixture. The $MgAl_2O_4$ sheet is too thin so as to approximate that its deformation is equal to the deformation of PMMA resin. The assembly is placed under the Raman laser with different deformation of 0 μm, 50 μm, 80 μm, 100 μm and 150 μm applied to it respectively, and the stress on $MgAl_2O_4$ ceramic can be calculated by four-point bending formula, and the corresponding peak shift under Raman spectrum can be obtained at the same time. The specific assembly method has been introduced in previous research and doesn't be repeated here [26].

The microstructure of the joint was characterized by SEM (Gemini560, Germany) with an equipped EDS (EDAX PV72-60030XVF). For the smaller phase in the joint, TEM (Talos F200x, FEI, USA) was be taken to characterize each phase. TEM sample was prepared by focused ion beam (FIB, TESCAN SOLARIS, Czech Republic). Through accurate cutting, transferring and thinning of samples, ultra-thin sample suitable for transmission electron microscope analysis is finally obtained. XRD (D8 Advanced, Bruker, Germany) was utilized to characterize the phases in the interface. The morphologies of joint and were observed by ultra-depth optical microscope (VHX-1000E, KEYENCE, Japan). The shear strength of the joints was tested by electronic universal testing machine (AG-X Plus, Instron, USA), as shown in Fig. 1c. The loading rate was set to 0.5 mm/s and at least 3 specimens were brought in the test to pledge the reliability of the experimental results. Additionally, the shear strength is calculated used the area of the whole welding region, referring to existing studies [27,28], and the direction of the shear test is perpendicular to the welding line with the reason given in Fig. S1. Raman spectroscopy system with the 532 nm line of an Ar-ion laser (InVia-Reflex, Renishaw, UK) was employed to characterize the Raman peak shift of the sample, which provided the basis for the follow-up calculation of residual stress in the joint.

3. Result and discussion

In our previous experiments, the joint with 100 mW/0.25 mm/s has the best mechanical properties (Fig. S2), so this parameter is used in this paper to conduct the follow-up experiments. The typical microstructure of $MgAl_2O_4$/Ti6Al4V joint obtained by femtosecond laser welding at 100 mW/0.25 mm/s with the hatching distance of 0.05 mm and scanning path of concentric circle is shown in Fig. 2. It can be seen that the joint is well welded with no obvious cracks. At the interface between $MgAl_2O_4$ and Ti6Al4V, some mountain-like structures with similar morphology distributed evenly, which used to be called welding zone. This unique structure caused by femtosecond laser welding has anchoring effect and forms a mechanical interlocking structure at the interface, which is of great benefit to the improvement of joint strength and the reduction of the residual stress in the joint according to previous FE models [20,21,29,30]. There are two different phases with bright and dark contrast mixed in the welding zone and some microvoids are superjacent above it, which may be caused by the melting of $MgAl_2O_4$ ceramic to fill the gap under the action of laser [[31], [32], [33]].

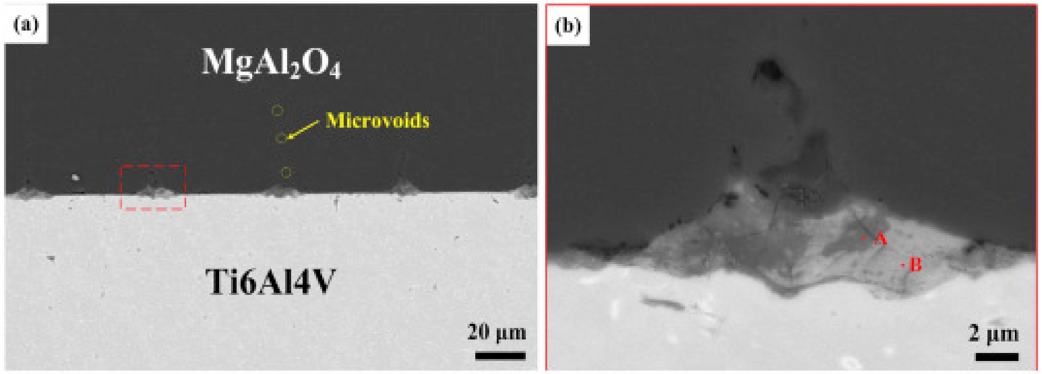

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Fig. 2. (a) Typical microstructure of $MgAl_2O_4$/Ti6Al4V joint welded by femtosecond laser at 100 mW/0.25 mm/s, (b) Magnified figure of the region in (a).

According to line scanning and mapping results in Fig. 3, the distribution of elements in the welding zone can be visible clearly. Five elements, which are O, Mg, Al, Ti, V, are mixed together in the welding zone with distribution gradient change. In the bright phase B, the contents of Ti and V increased obviously. On the contrary, their contents are dropping sharply in the dark phase A while Mg, Al and O are mainly distributed.

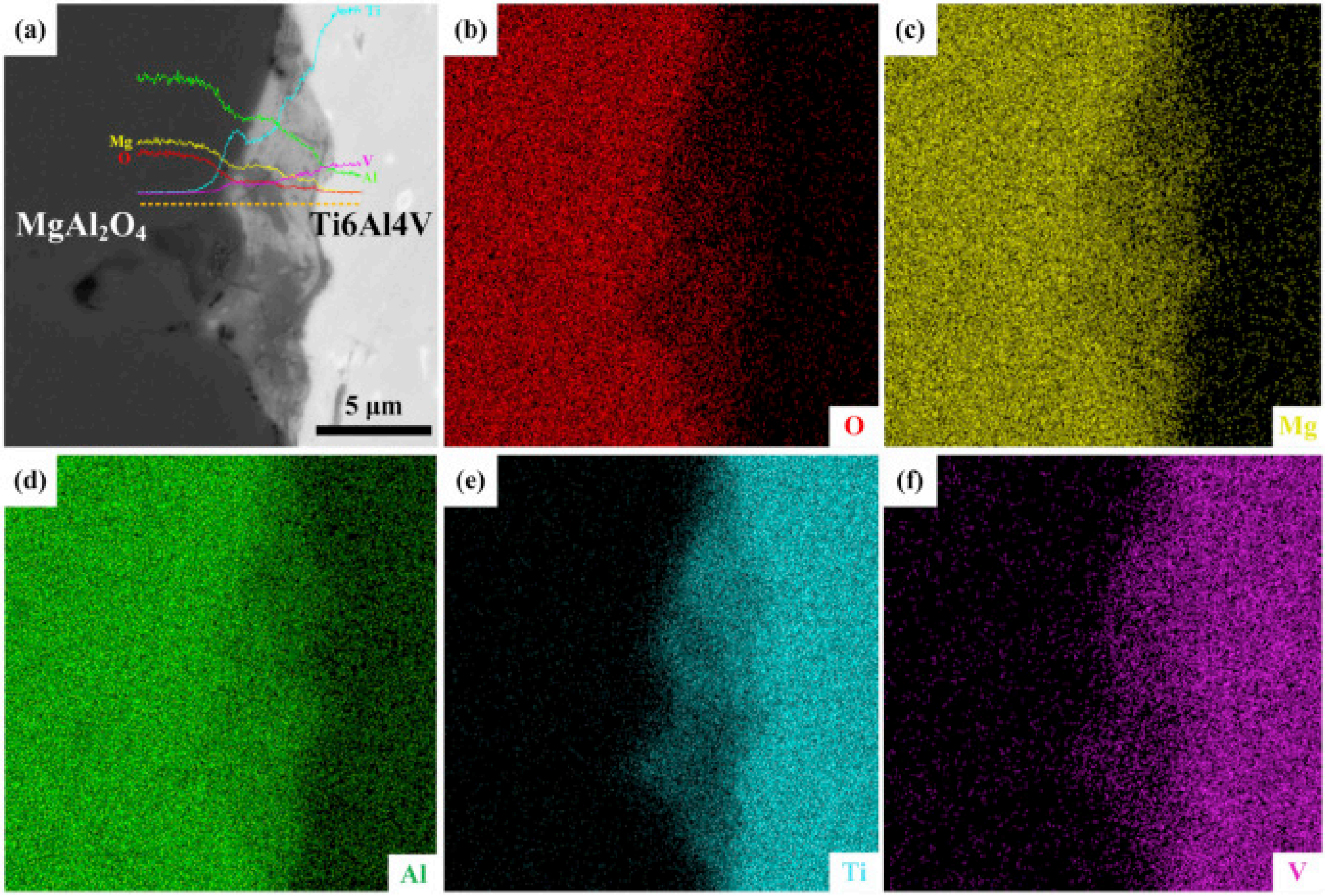

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Fig. 3. (a) The BSE image of the joint with line scanning, the element distribution measured via EPMA of (b) O, (c) Mg, (d) Al, (e) Ti and (f) V.

Further enlarge one of the welding zones in Fig. 2a and adopt EDS analysis to clarify the elemental composition of two phases with different contrast. The following conclusions can be drawn. The dark phase A is mainly composed of Mg, Al, O with the ratio of 1:2:4, so it can be confirmed as mainly $MgAl_2O_4$. At the same time, the content of Ti element cannot be ignored. Femtosecond laser has ultra-high peak power density, which can cause a series of nonlinear absorption effects of transparent $MgAl_2O_4$ ceramic.

It is reasonable to speculate that Ti element is likely to be oxidized to produce Ti oxide at high temperature. But in the dark phase A, the content of Ti_xO_y is relatively low. In the bright phase B, the content of Ti is obviously increased with a small distribution of Mg and Al, so it is probably a mixture of $MgAl_2O_4$ and Ti_xO_y. The relevant details are listed in Table 1.

Table 1. The EDS analysis results of the various phases in the joint shown in Fig. 2b (at. %).

position	**O**	**Mg**	**Al**	**Ti**	**V**	**Possible phase**
A	48.14	10.30	26.43	14.54	0.59	$MgAl_2O_4$+Ti_xO_y (low-content)
B	36.74	7.27	20.50	34.03	1.46	$MgAl_2O_4$+Ti_xO_y

In order to further confirm the phase in the welding zone for its small size, TEM analysis is adopted and the results obtained are shown in Fig. 4. It can be seen that dark phase A and bright phase B are dispersed in the welding zone and their shapes vary in size, which caused by the rapidity of femtosecond laser. The dark phase is rich in Mg, Al and O, which can be judged as $MgAl_2O_4$ by combining its SAED pattern. The light-colored phase is mainly composed of Ti and it can be determined as Ti_2O, the formation of which can be explained that the extremely high temperature in the focal region due to the ultra-high peak density of femtosecond laser makes Ti bound to be oxidized in the air [34,35]. XRD characterization method has been also adopted to confirm this result, which is shown in Fig. 5.

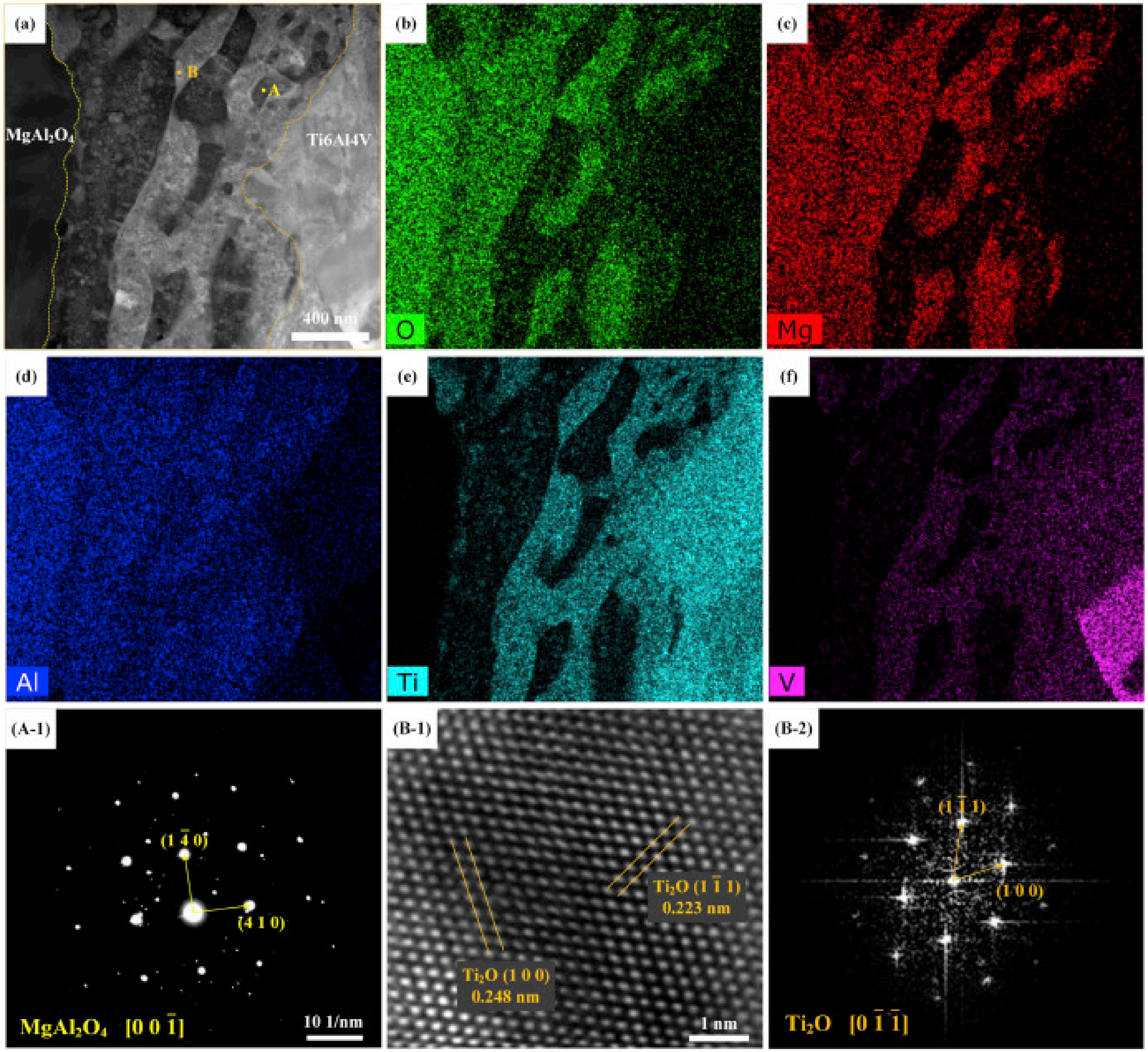

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Fig. 4. TEM image of the $MgAl_2O_4$/Ti6Al4V interface: (a) the HADDF image, (b–f) element distributions of O, Mg, Al, Ti and V, (A-1) SAED pattern of $MgAl_2O_4$, (B-1) the HRTEM image of Ti_2O, (B-2) the diffraction pattern after FFT of (B-1).

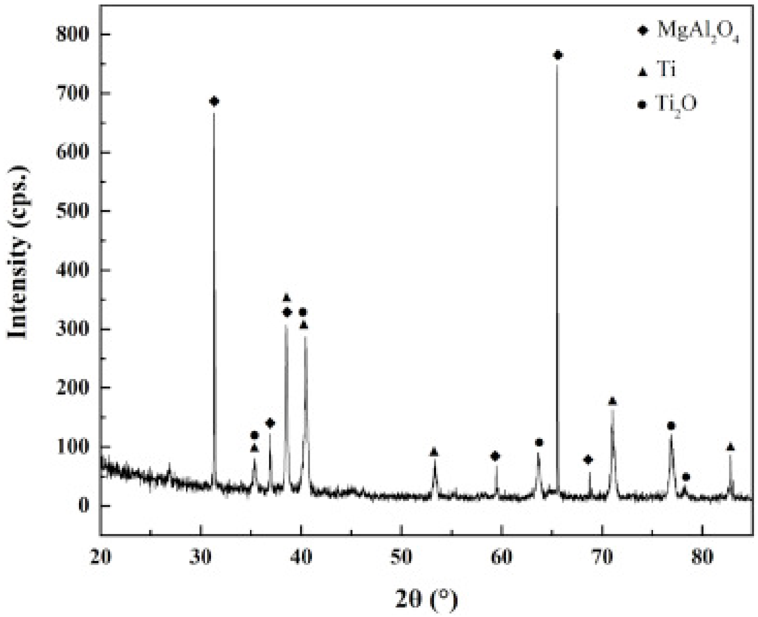

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Fig. 5. The phases of $MgAl_2O_4$/Ti6Al4V joint tested by XRD.

Different from brazing and diffusion welding, femtosecond laser welding has three-dimensional spatial selectivity [36], which allows the effective joining area and shape of the joint to be controlled. Ergo, it is necessary to explore the influence of scanning path on the joint. Fixed laser power at 100 mW, welding speed at 0.25 mm/s and hatching distance at 0.05 mm, three different scanning paths, which are serpentine, fillet-serpentine and concentric circle, are adopted respectively to join $MgAl_2O_4$ and Ti6Al4V. Fig. 6 shows schematic diagrams and the corresponding macroscopic morphologies of three different processing modes separately. The dark regions in Fig. 6a_1-c_1 present the actual joining area between $MgAl_2O_4$ and Ti6Al4V. The white areas between the welding lines are the space of the laser scanning, that is, the unbonded areas. It is worth mentioning that the laser parameters which govern the laser energy released onto the workpiece per unit area and time like laser power and scanning speed are constant, so the microstructure of the joint attained by different scanning paths will not change theoretically. Therefore, microstructure analysis is not conducted here, but only pay attention to the influence of scanning path on the mechanical properties of the joint.

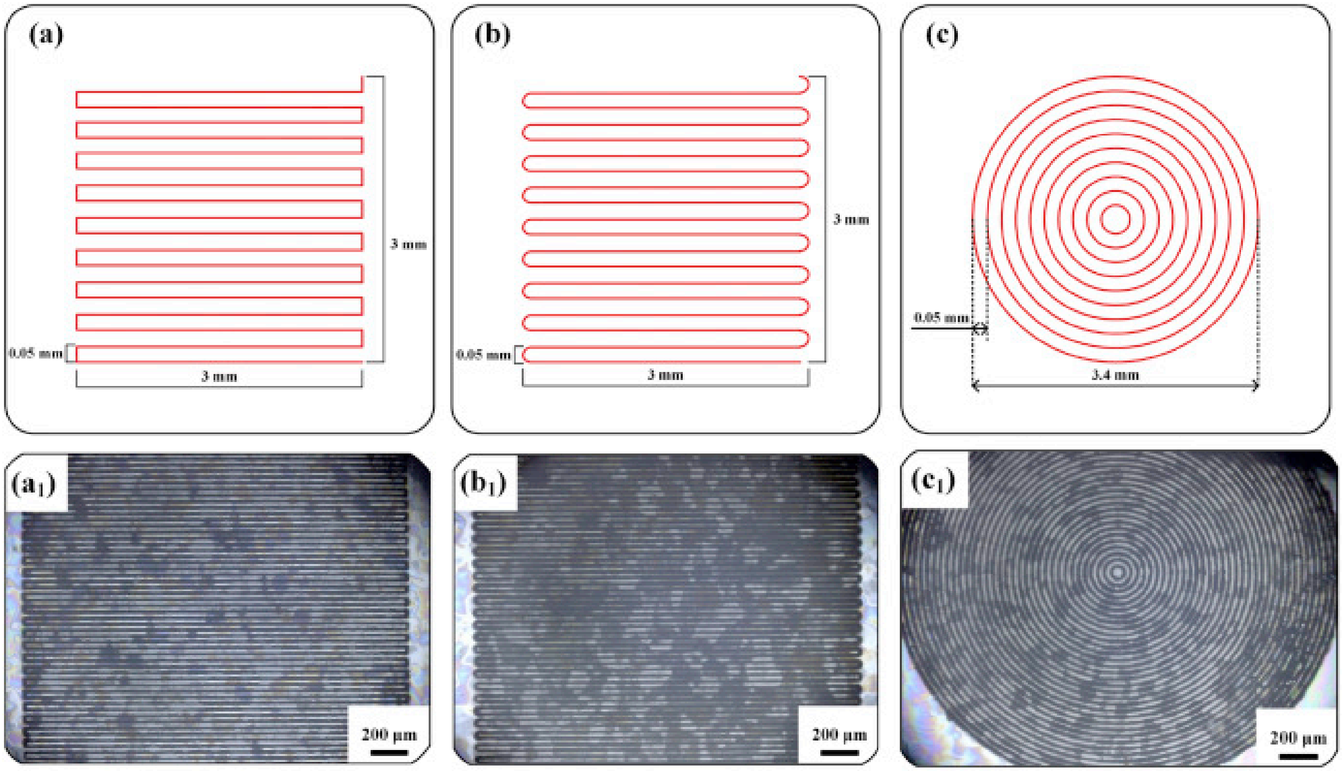

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Fig. 6. schematic diagrams and the corresponding macroscopic morphologies of three scanning paths: (a-a_1) serpentine, (b-b_1) fillet-serpentine and (c-c_1) concentric circle.

Fig. 7 presents the fracture morphologies and the results of room-temperature shear strength testing of the joints. It can be seen that the cracks all start from the interface and extend into the ceramic, with some ceramic fragments remaining at the welding line. The shell-like and step-like morphologies indicate the brittle fracture characteristics of the joints. The joint with the shape of concentric circle has the maximum strength of 23.35 MPa, followed by the fillet-serpentine-shape joint, and the serpentine joint has the lowest strength. The welding line density of serpentine, fillet-serpentine and concentric circle samples are 20.3 mm/mm^2, 20.2 mm/mm^2 and 20.6 mm/mm^2, respectively. The small difference between them will not have a great effect on the shear strength. So, the result of shear strength may be caused by the following reasons. There is some difference in thermal expansion coefficient between $MgAl_2O_4$ and Ti6Al4V (CTE of $MgAl_2O_4$: 7.6×10^{-6}/°C [37], CTE of Ti6Al4V: $8.5–10\times10^{-6}$/°C [38]), which leads to the generation of residual stress during the process of post-weld cooling. However, for brittle ceramics, residual stress causes cracks to initiate and spread in this, which leads to the failure of the joint [39]. In many finite element simulation studies, it can be known that there will be stress concentration at the ceramic corner [40,41]. In this study, the serpentine welding line will also cause corner similar to brazing, where the stress concentration will form and ultimately lead to the decrease of joint strength. After changing the right-angled corner into rounded corner, the stress may be relieved to some extent. When the scanning path is concentric circle, the existence of no corner leads to relatively uniform stress distribution, and the strength of the joint reaches the highest at this time.

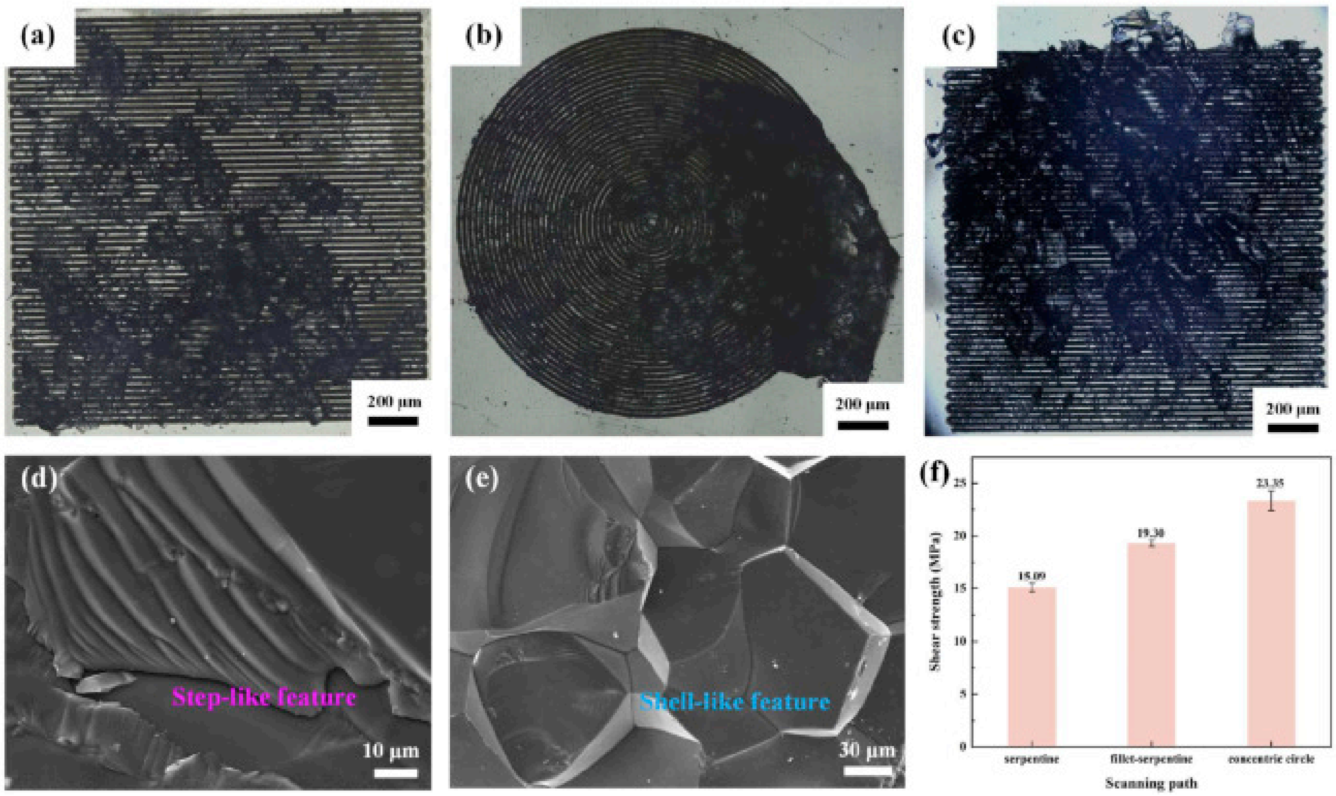

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Fig. 7. The macroscopic morphologies of the joint fracture surfaces near the Ti6Al4V side with the scanning path of (a) serpentine, (b) concentric circle and (c) fillet-serpentine; typical fracture morphology near Ti6Al4V side: (d) step-like morphology, (e) shell-like morphology; (f) the shear strength of $MgAl_2O_4$/Ti6Al4V joints attained with different scanning paths.

In order to prove the above conclusion, the residual stress of the joint was measured. Residual stress is the most common problem in the ceramic/metal joint, and its existence will seriously reduce the life of the joint and lead to its failure. For example, the residual tensile stress will accelerate the microcrack propagation at the interface between ceramics and metals, and reduce the joint strength. In addition, it will reduce the fracture toughness of the material and cause the joint brittle fracture under the load lower than the theoretical strength. The shape change of the welding area will form local stress concentration, which will further reduce fatigue performance [42,43]. Therefore, it is very practical to measure the residual stress in the joint. $MgAl_2O_4$ is a transparent to Raman, which makes Raman spectroscopy become the best method in this study. Residual stress in the transparent ceramic/metal joints have been measured by Raman spectroscopy in many researches [26,44], which is confirmed as consequential and reliable. The principle of measuring residual stress by Raman spectroscopy is that when residual stress inside of the sample, some peaks sensitive to it will shift and the change of Raman peak frequency offset has a corresponding relationship with the stress, which is usually proportional. Therefore, the relationship between Raman peak shift and stress in materials can be established as Eq. (1)

$$\Delta\omega = \omega_{\mathrm{i}} - \omega_0 \approx \sigma \cdot \Pi \qquad (1)$$

where $\Delta\omega$ is Raman peak frequency shift, ω_0 and ω_{i} present Raman peak position in stress-free state and in stressed state, respectively. σ stands for the stress and Π is the piezoelectric coefficient of $MgAl_2O_4$ in this study.

The commercial origin data processing software is used to fit the deviation of Raman peak under different deformation. The Raman spectra of $MgAl_2O_4$ ceramic ranges from 200 cm^{-1} to 800 cm^{-1} and the peak position lied in ~300 cm^{-1} was chosen to fit since it is a single peak [45,46], and the results are shown in Fig. 8a. Draw the Raman peak shift under different deformation calculated by Eq. (1) and the corresponding stress into a scatter plot and a straight-line fitting method is adopted to obtain the value of Π of −0.01555 cm^{-1}/MPa.

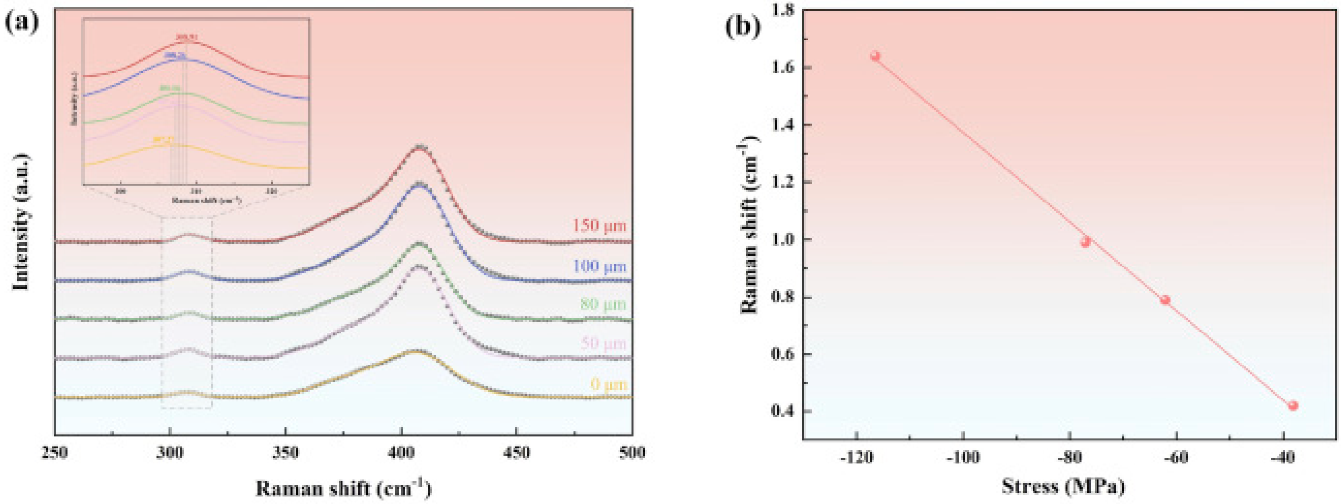

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Fig. 8. (a) The fitted results of $MgAl_2O_4$ Raman spectrum data with 0, 50, 80, 100, 150 μm deformation, (b) Raman frequency shift-stress scatter plot fitting.

So far, the Raman frequency shift-stress conversion coefficient Π has been obtained, and the corresponding stress value can be calculated by Eq. (1) only by obtaining the Raman peak offset of the joints with different scanning paths. First, the laser is focused on the center of the surface on the ceramic side of the joint. And then, the laser focus was moved downward straightly into the ceramic until it can be clearly focused at the interface, which profits from the transparency of $MgAl_2O_4$ ceramic to the laser. $MgAl_2O_4$ ceramic is poly-crystalline, so the stress at the center point is symmetrically distributed and the deformation in z direction is the smallest, which can be regarded as a biaxial stress state approximately. Hence, the average stress value of the joint can be calculated. Fitting the Raman peaks obtained from the joints with different scanning paths, the obtained Raman peak shift relative to the stress-free state is brought into Eq. (1), and the calculated stress values are −33.76 MPa, −32.15 MPa and −22.83 MPa, respectively.

From the above results (Fig. 9), it can be seen that the joint residual stress of concentric circle welding line is the smallest, and the variation tendency of joint residual stress and the joint strength echoes to a great extent. That is to say, the circular scanning path reduces the stress concentration in the joint and can distribute the mechanical constraints more evenly under the action of external force, which thanks to its less starting and stopping points, it can limit crack initiation from corners [47]. Therefore, the stress in the joint with concentric circle scanning path is the smallest and the shear strength is also the highest.

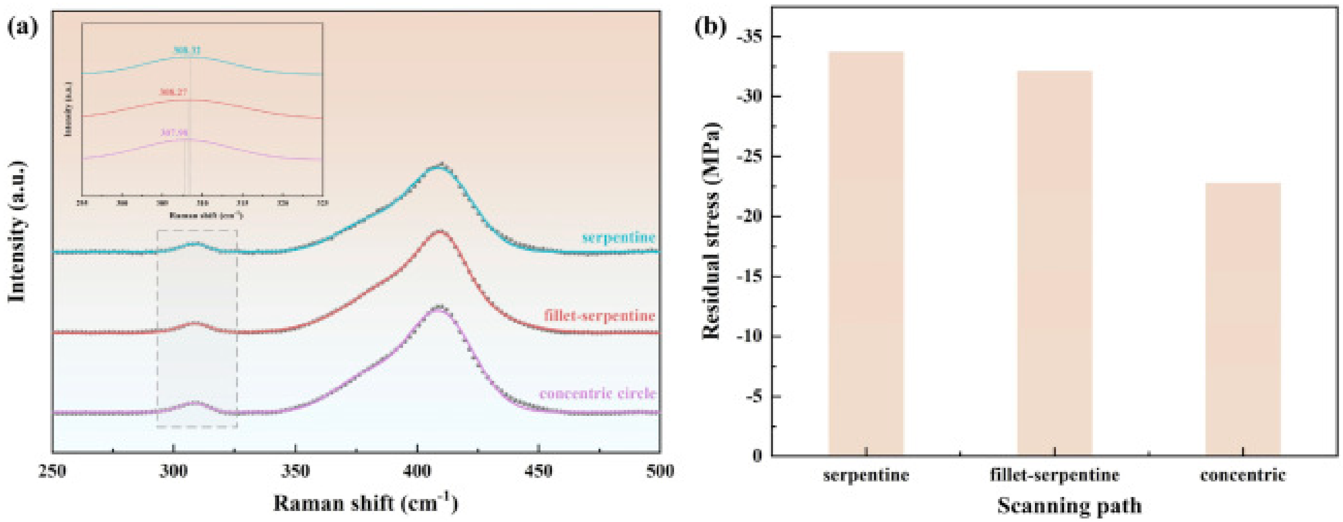

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Fig. 9. The figures of $MgAl_2O_4$/Ti6Al4V joints with different scanning paths: (a) the fitted results of Raman spectrum data, (b) the residual stress values.

Hatching distance is an important parameter in femtosecond laser welding, which is defined here as the distance between consecutive welding lines (hereinafter referred to as h). Fig. 10 displays the macroscopic morphologies of $MgAl_2O_4$/Ti6Al4V joint obtained at 100 mW/0.25 mm/s with different hatching distances (0.01 mm, 0.025 mm, 0.05 mm, 0.075 mm and 0.1 mm). It can be seen that when h=0.01 mm, the welding lines overlapped with each other and are difficult to distinguish. With the increase of hatching distance, each welding line becomes legible. Hatching distance has little effect on the width of welding lines, and Newton's rings are clearly visible outside the bonding area (Fig. 10a_1), which manifests that the joining area irradiated by femtosecond laser is tight and sound.

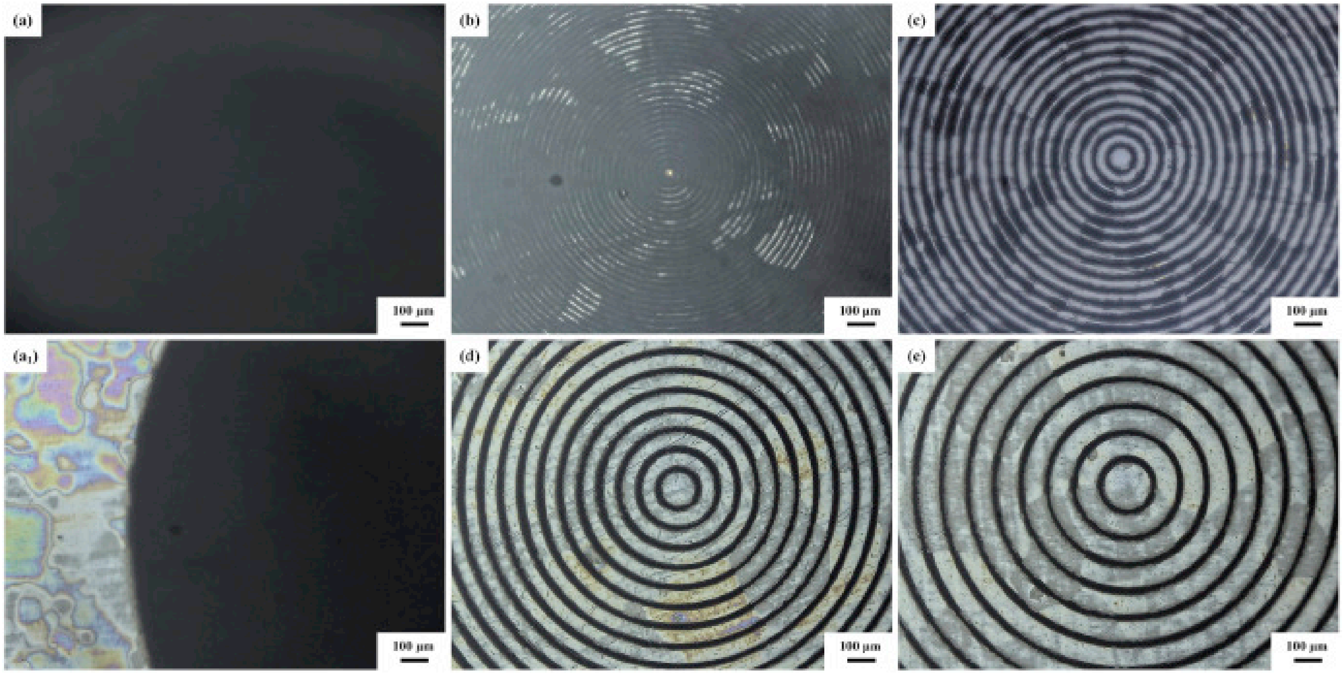

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Fig. 10. The macroscopic morphologies of $MgAl_2O_4$/Ti6Al4V joint with different hatching distances: (a) 0.01 mm, (a_1) supplementary diagram of (a), (b) 0.025 mm, (c) 0.05 mm, (d) 0.075 mm and (e) 0.1 mm.

Fig. 11 shows the interface microstructure of $MgAl_2O_4$/Ti6Al4V joint with different hatching distance. It can be seen that the size and morphology of the welding spots are almost similar regardless of the hatching distance. When h=0.01 mm, the welding spots are closely packed together and overlapped with each other partly. When h=0.025 mm and 0.05 mm, the whole interface is combined tightly even in the unjoined area. As the hatching distance continues to increase, cracks begin to appear in the $MgAl_2O_4$ ceramic of the joint.

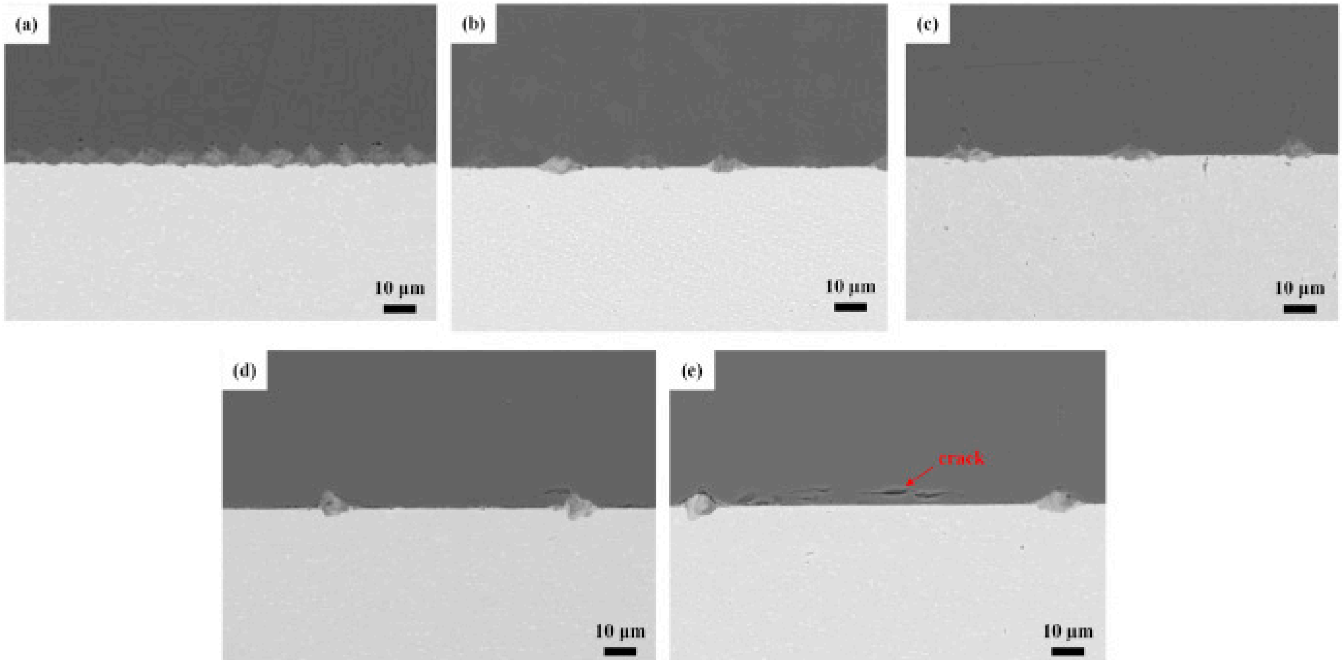

Fig. 11. The interfacial microstructure of $MgAl_2O_4$/Ti6Al4V joint with different hatching distance: (a) 0.01 mm, (b) 0.025 mm, (c) 0.05 mm, (d) 0.075 mm and (e) 0.1 mm.

The shear strength of joints obtained at different hatching distance was tested with the result shown in Fig. 12a. With the increase of hatching distance, the shear strength of the joint first increases and then decreases. When h=0.025 mm, the joint has the highest shear strength of 27.56 MPa. It is not difficult to understand that the strength of the joint decreases when the hatching distance is too long because of the difficulty on gap filling fully of the molten material generated by laser. Evidence can also be obtained from the fracture morphologies, because the $MgAl_2O_4$ ceramic residue at the welding line decreases gradually with the increase of hatching distance, which shows that the joining between $MgAl_2O_4$ ceramic and Ti6Al4V alloy is not tight. However, the joint strength does not increase continuously with the decrease of hatching distance. Here, this result can be attributed to the fracture of the anchoring between $MgAl_2O_4$ and Ti6Al4V due to the laser scanning of the processed line [48]. In addition, the excessive accumulation of heat will also lead to the initiation of cracks in ceramics under thermal stress, which leads to the fracture of ceramic under external force [42,49].

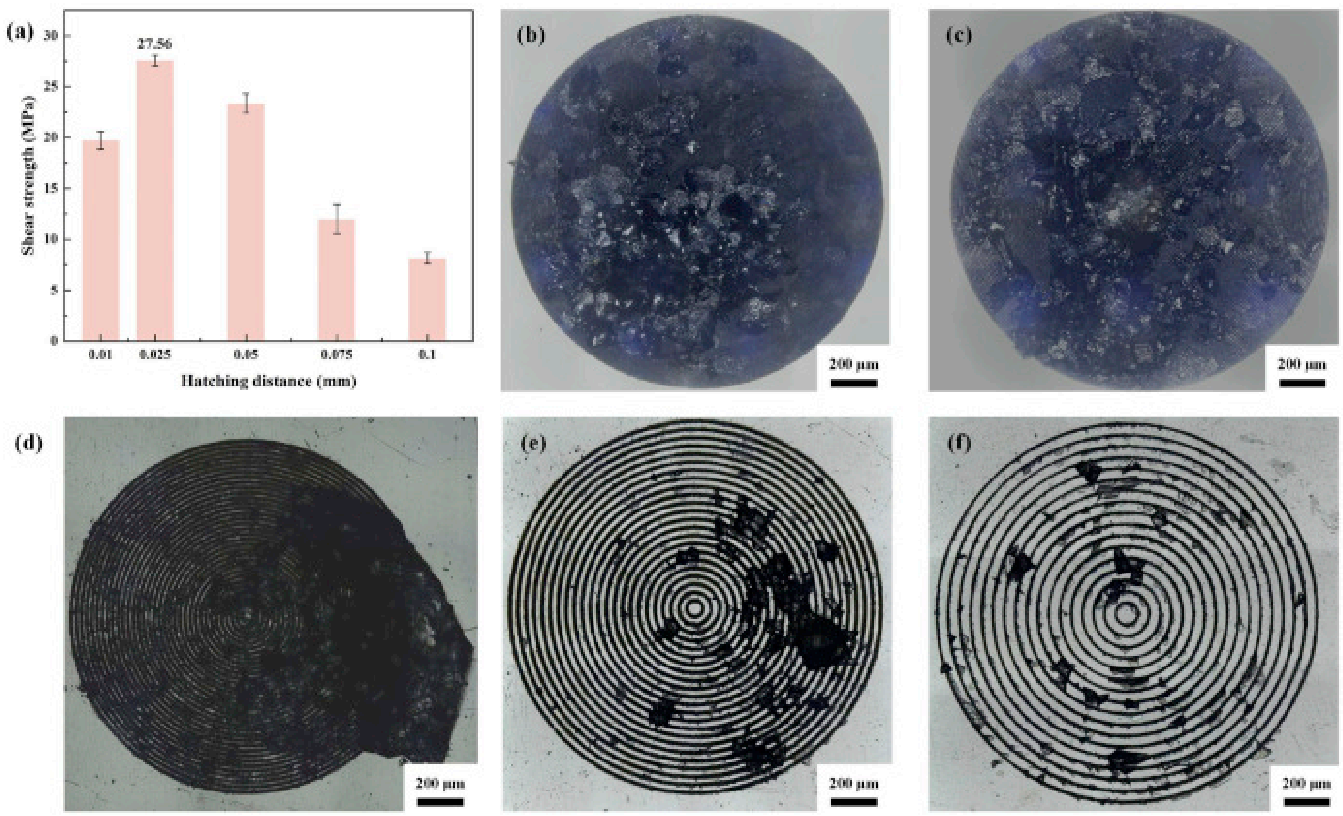

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Fig. 12. (a) The shear strength of the $MgAl_2O_4$/Ti6Al4V joints attained at 100 mW/0.25 mm/s with different hatching distance; the macroscopic morphologies of the joint fracture surfaces near the Ti6Al4V side with different hatching distance: (b) 0.01 mm, (c) 0.025 mm, (d) 0.05 mm, (e) 0.075 mm and (f) 0.1 mm.

Fig. 13 shows the residual stress of $MgAl_2O_4$/Ti6Al4V joint obtained at 100 mW/0.25 mm/s with different hatching distances (0.01 mm, 0.025 mm, 0.05 mm, 0.075 mm and 0.1 mm). It can be seen that with the increase of hatching distance, the residual stress of the joint tends to decrease. It can be explained as follows. The residual stress of the joint is caused by the mismatch of

CTE between $MgAl_2O_4$ ceramic and Ti6Al4V alloy. However, with the increase of hatching distance, the real joining area of the joint decreases, which results in the reduction of residual stress in the joint. Therefore, when h=0.1 mm, the residual stress of the joint has the smallest value of −19.61 MPa. Although the residual stress of the joint is the smallest when h=0.1 mm, the strength of the joint is not the highest. It can be seen that there is not a simple corresponding relationship between the residual stress and the shear strength of the joint, that is, the smaller the residual stress of the joint, the higher the joint strength. The strength of the joint was affected by both the real joining area and the residual stress.

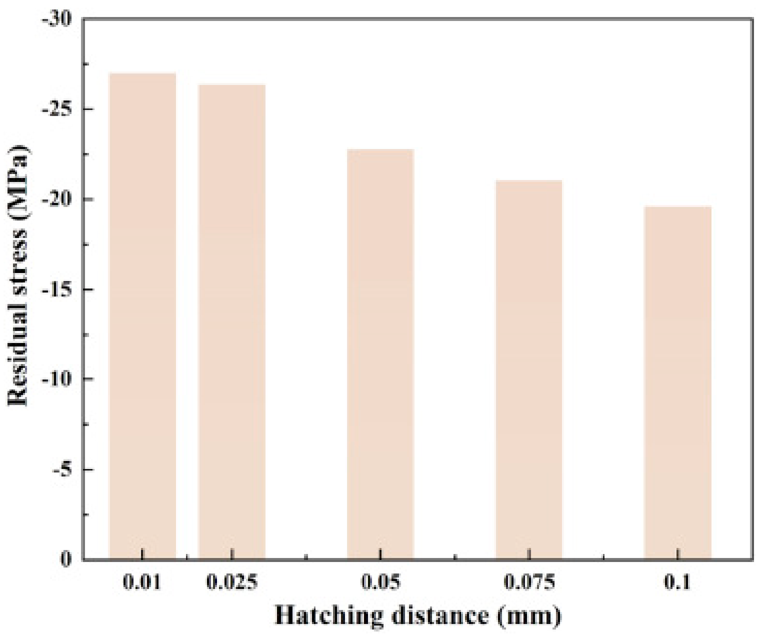

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Fig. 13. Residual stress of $MgAl_2O_4$/Ti6Al4V joints with different hatching distance.

4. Conclusion

In this study, the direct joining between $MgAl_2O_4$ and Ti6Al4V was successfully achieved by femtosecond laser welding. The microstructure of the welding zone was characterized in the general. Meanwhile, the effect of hatching distance and scanning path on the microstructure and mechanical properties of $MgAl_2O_4$/Ti6Al4V joint was clarified determinedly, which is rarely studied in previous researches. The conclusions are drawn as follows.

(1) Under the action of femtosecond laser with high peak intensity, the good joining between $MgAl_2O_4$ and Ti6Al4V has been achieved and three phases of $MgAl_2O_4$, Ti_2O and Ti bave been found in the joint.

(2) Combined with Raman spectroscopy, the residual stress of the joint is measured and it is concluded that the residual stress of joint with concentric circle welding line is the smallest and the joint strength is the highest.

(3) With the increase of hatching distance, the shear strength of the joint first increases and then decreases. The highest shear strength of the joint is 27.56 MPa, which is welded with the concentric circle scanning path of 0.025 mm hatching distance.

CRediT authorship contribution statement

Hao Jiang: Writing – review & editing, Writing – original draft, Validation, Methodology. **Chun Li:** Writing – review & editing, Software, Data curation. **Xiaojian Mao:** Methodology, Funding acquisition, Data curation. **Wendi Zhao:** Software, Funding acquisition, Conceptualization. **Ningce Wei:** Supervision, Resources, Conceptualization. **Xiaoqing Si:** Writing – review & editing, Methodology, Investigation. **Junlei Qi:** Validation, Resources, Investigation. **Jian Cao:** Writing – review & editing, Validation, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is the Supplementary data to this article.

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Multimedia component 1.

Recommended articles

Data availability

Data will be made available on request.

References

[1] J. Chen, W. Shang, S. Xu
Endoir: a GAN-Based method for fiber bundle endoscope image restoration
Opt Laser. Eng., 184 (2025)
Google Scholar ↗

[2] L.M. Gherasie, C. Voiosu, R. Bartel, R. Hainarosie, I.G. Ionita, M.D. Zica, V. Zainea
Techniques for otosclerosis surgery: ear surgery from the microscope to the endoscope - a literature review
Journal of Otology (2024)
Google Scholar ↗

[3] R. Liu, Q. Liu, F. Ye, Q. Pang, H. Tang, K. Yi, K. Zhao, G. Sun
Endoscope-assisted keyhole surgery for traumatic subacute subdural hematoma evacuation in emergencies
World Neurosurg., 172 (2023), pp. e194-e200
View PDF View article View in Scopus ↗ Google Scholar ↗

[4] C.M. Lim, G.N. Rao, H. Ren

Real time in-vivo miniature endoscopic surveillance system for imaging of nasopharynx
World J. Otorhinolaryngol. Head Neck Surg., 6 (1) (2020), pp. 4-9
View PDF View article Crossref ↗ View in Scopus ↗ Google Scholar ↗

[5] S.Y. Kim, Y.J. Park, H.J. Ma, J.-W. Ko, J.W. Lee, S.H. Bae, H.N. Kim
Optimization of Ca additive and HIP condition in fabrication of the transparent $MgAl_2O_4$ ceramics
Int. J. Refract. Metals Hard Mater., 120 (2024)
Google Scholar ↗

[6] Y. Ren, W. Zhu, Y. Yu, H. Zou, G. Zu, Y. Han, X. Ran
Fabrication of sapphire/$MgAl_2O_4$/sapphire laminated transparent composite by brazing with BaO-B_2O_3-SiO_2 glass filler
Ceram. Int., 50 (18) (2024), pp. 32295-32305
View PDF View article View in Scopus ↗ Google Scholar ↗

[7] J.T. Fan, F.F. Wu, Z.F. Zhang, F. Jiang, J. Sun, S.X. Mao
Effect of microstructures on the compressive deformation and fracture behaviors of $Zr_{47}Cu_{46}Al_7$ bulk metallic glass composites
J. Non-Cryst. Solids, 353 (52–54) (2007), pp. 4707-4717
View PDF View article View in Scopus ↗ Google Scholar ↗

[8] J. Fan, F. Wu, D. Li
Dynamic compressive response of a dendrite-reinforced Ti-based bulk metallic glass composite
Mater. Sci. Eng., 720 (2018), pp. 140-144
View PDF View article View in Scopus ↗ Google Scholar ↗

[9] J. Fan, T. Fu
Toughened austenitic stainless steel by surface severe plastic deformation
Mater. Sci. Eng., 552 (2012), pp. 359-363
View PDF View article View in Scopus ↗ Google Scholar ↗

[10] X.M. Zhang, T.M. Yue, H.C. Man
Enhancement of ceramic-to-metal adhesive bonding by excimer laser surface treatment
Mater. Lett., 30 (5) (1997), pp. 327-332
View PDF View article View in Scopus ↗ Google Scholar ↗

[11] H. Mei, D. Zhang, J. Xia, L. Cheng
Effect of heat treatment on the riveted joints of two-dimensional C/SiC composites
Compos. B Eng., 120 (2017), pp. 159-167
View PDF View article View in Scopus ↗ Google Scholar ↗

[12] J. Shin, A. Sharma, D.-h. Jung, J.P. Jung
Effect of Sn content on filler and bonding characteristics of active metal brazed Cu/Al_2O_3 joint
Korean Journal of Metals and Materials, 56 (5) (2018), pp. 366-374
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[13] F.A. Mir, N.Z. Khan, S. Parvez
Recent advances and development in joining ceramics to metals
Mater. Today Proc., 46 (2021), pp. 6570-6575
View PDF View article Google Scholar ↗

[14] H. Wang, B. Liu, D. Pang, M. Hu
Welding of glass and single crystal graphite film using a high repetition fs laser
J. Non-Cryst. Solids, 647 (2025)
Google Scholar ↗

[15] D. Tan, K.N. Sharafudeen, Y. Yue, J. Qiu
Femtosecond laser induced phenomena in transparent solid materials: fundamentals and applications
Prog. Mater. Sci., 76 (2016), pp. 154-228
View PDF View article View in Scopus ↗ Google Scholar ↗

[16] T. Tamaki, W. Watanabe, J. Nishii, K. Itoh

Welding of transparent materials using femtosecond laser pulses
Jpn. J. Appl. Phys., 44 (5L) (2005)
Google Scholar ↗

[17] J. Huo, Z. Zeng, J. Yuan, M. Luo, A. Luo, J. Li, H. Yang, N. Zhao, Q. Zhang
Welding between rough copper foil and silica glass using green femtosecond laser
Opt Laser. Technol., 181 (2025)
Google Scholar ↗

[18] I. Miyamoto, K. Cvecek, M. Schmidt
Simple evaluation procedure of nonlinear absorptivity in glass welding using USLP with high-pulse repetition rates
Phys. Procedia, 39 (2012), pp. 526-533
View PDF View article View in Scopus ↗ Google Scholar ↗

[19] Y.M. Ho, C.H. Lee, J.R. Ho, C.K. Lin, P.C. Tung, Y.-S. Lee
Assessment of glass-to-glass welding by USP lasers with machine learning approaches
Manufacturing Letters, 35 (2023), pp. 160-165
View PDF View article View in Scopus ↗ Google Scholar ↗

[20] T.O. Lipateva, A.G. Okhrimchuk, A.S. Lipatiev, I.A. Karateev, S.S. Fedotov, S.V. Lotarev, A.V. Shestakov, V.N. Sigaev
Robust and adhesive-free joint of Nd:YAG crystals by femtosecond laser-assisted welding
Opt Laser. Technol., 146 (2022)
Google Scholar ↗

[21] R. Pan, D. Yang, T. Zhou, Y. Feng, Z. Dong, Z. Yan, P. Li, J. Yang, S. Chen
Micro-welding of sapphire and metal by femtosecond laser
Ceram. Int., 49 (13) (2023), pp. 21384-21392
View PDF View article View in Scopus ↗ Google Scholar ↗

[22] S.S. Fedotov, T.O. Lipat'eva, A.S. Lipat'ev, G.Y. Shakhgil'dyan, S.V. Lotarev, V.I. Savinkov, V.N. Sigaev

Femtosecond laser welding of glass and sitall with substantially different values of the LTEC
Glass Ceram., 80 (1–2) (2023), pp. 41-44
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[23] I. Miyamoto, K. Cvecek, Y. Okamoto, M. Schmidt, H. Helvajian
Characteristics of laser absorption and welding in FOTURAN glass by ultrashort laser pulses
Opt. Express, 19 (23) (2011), pp. 22961-22973
View in Scopus ↗ Google Scholar ↗

[24] I. Miyamoto, K. Cvecek, M. Schmidt
Crack-free conditions in welding of glass by ultrashort laser pulse
Opt. Express, 21 (12) (2013), pp. 14291-14302
View in Scopus ↗ Google Scholar ↗

[25] K. Cvecek, F. Stenglein, I. Miyamoto, M. Schmidt
Influence of wavelength on glass welding by ultra-short laser pulses
Journal of Laser Micro Nanoengineering, 12 (2) (2017), pp. 115-119
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[26] H. Jiang, C. Li, X. Mao, L. Chen, W. Zhao, X. Si, J. Qi, Z. Dong, J. Cao
Vacuum brazing of AlON and Ti_2AlNb with $LiAlSiO_4$ enhanced Ag-Cu-Ti composite fillers: microstructure, mechanical properties and measurement of residual stress
Mater. Sci. Eng., 846 (2022)
Google Scholar ↗

[27] T. Tamaki, J. Neev, W. Watanabe, S. Nolte, A. Heisterkamp, K. Itoh, C.B. Schaffer
Laser micro-welding of silicon and borosilicate glass using nonlinear absorption effect induced by 1558-nm femtosecond fiber laser pulses
Commercial and Biomedical Applications of Ultrafast Lasers VII (2007)
Google Scholar ↗

[28] J. Zhan, Y. Gao, J. Sun, W. Zhu, S. Wang, L. Jiang, X. Li
Mechanism and optimization of femtosecond laser welding fused silica and aluminum
Appl. Surf. Sci., 640 (2023)
Google Scholar ↗

[29] Y. Ozeki, T. Inoue, T. Tamaki, H. Yamaguchi, S. Onda, W. Watanabe, T. Sano, S. Nishiuchi, A. Hirose, K. Itoh
Direct welding between copper and glass substrates with femtosecond laser pulses
APEX, 1 (2008)
Google Scholar ↗

[30] I. Miyamoto, K. Cvecek, Y. Okamoto, M. Schmidt
Internal modification of glass by ultrashort laser pulse and its application to microwelding
Appl. Phys. A, 114 (1) (2013), pp. 187-208
Crossref ↗ Google Scholar ↗

[31] H. Yu, J.X. Zhao, L.J. Zhang, S.-J. Na
Femtosecond laser welding of sapphire-copper using a thin film titanium interlayer
Opt Laser. Technol., 177 (2024)
Google Scholar ↗

[32] M. Kahle, D. Nodop
Glass welding with ultra-short laser pulses and long focal lengths A process model
Proced. CIRP, 111 (2022), pp. 617-620
View PDF View article View in Scopus ↗ Google Scholar ↗

[33] J. Fan, T. Fu
Composite design of thin hard $AlNi_3$ coating on soft stainless steel for making the improved impact resistance
Surf. Coating. Technol., 368 (2019), pp. 1-7
View PDF View article View in Scopus ↗ Google Scholar ↗

[34] J.A. Sandoval-Robles, C.A. Rodriguez, E. Garcia-Lopez
Laser surface texturing and electropolishing of CoCr and Ti6Al4V-ELI alloys for biomedical applications
Materials, 13 (22) (2020)
Google Scholar ↗

[35] C.G. Moura, O. Carvalho, L.M.V. Goncalves, M.F. Cerqueira, R. Nascimento, F. Silva
Laser surface texturing of Ti-6Al-4V by nanosecond laser: surface characterization, Ti-oxide layer analysis and its electrical insulation performance
Mater. Sci. Eng., C, 104 (2019), Article 109901
View PDF View article View in Scopus ↗ Google Scholar ↗

[36] X. Wang, H. Yu, P. Li, Y. Zhang, Y. Wen, Y. Qiu, Z. Liu, Y. Li, L. Liu
Femtosecond laser-based processing methods and their applications in optical device manufacturing: a review
Opt Laser. Technol., 135 (2021)
Google Scholar ↗

[37] Y. Ren, W. Zhu, S. Pu, Y. Chen, Y. Shen, Y. Han, X. Ran
Joining $MgAl_2O_4$ ceramics using ZnO-Al_2O_3-SiO_2 glass ceramic with precipitated $ZnAl_2O_4$
Ceram. Int., 49 (20) (2023), pp. 32835-32842
View PDF View article View in Scopus ↗ Google Scholar ↗

[38] Q. Li, Z. Liu, X. Liu, L. Wang, C. Zang, L. Li, P.E.J. Rivera-Diaz-del-Castillo
Enabling invar to Ti6Al4V transitions through copper in functionally graded laser powder bed fused components
J. Mater. Res. Technol., 32 (2024), pp. 2595-2608
View PDF View article View in Scopus ↗ Google Scholar ↗

[39] J. Fan, L. Zhu, J. Lu, T. Fu, A. Chen
Theory of designing the gradient microstructured metals for overcoming strength-ductility trade-off
Scr. Mater., 184 (2020), pp. 41-45

View PDF View article Google Scholar ↗

[40] T. Wang, J. Zhang, W. Lee, T. Ivas, C. Leinenbach
Numerical analysis on the residual stress distribution and its influence factor analysis for Si_3N_4/42CrMo brazed joint
Simulat. Model. Pract. Theor., 95 (2019), pp. 49-59
View PDF View article Google Scholar ↗

[41] Y. Wang, M. Liu, H. Zhang, Z. Wen, M. Chang, G. Feng, D. Deng
Fabrication of reliable ZTA composite/Ti6Al4V alloy joints via vacuum brazing method: microstructural evolution, mechanical properties and residual stress prediction
J. Eur. Ceram. Soc., 41 (7) (2021), pp. 4273-4283
View PDF View article View in Scopus ↗ Google Scholar ↗

[42] J.T. Fan, Y.M. Yan
Gradient microstructure with martensitic transformation for developing a large-size metallic alloy with enhanced mechanical properties
Mater. Des., 143 (2018), pp. 20-26
View PDF View article View in Scopus ↗ Google Scholar ↗

[43] X. Nai, S. Zhao, P. Wang, H.Y. Chen, P.C. Wang, X.G. Song, A. Vairis, W.Y. Li
Solid-state transformation composite filler: a new strategy to strengthen ceramic/metal joint
Mater. Des., 243 (2024)
Google Scholar ↗

[44] C. Li, X. Si, L. Chen, J. Qi, Z. Liu, Y. Huang, Z. Dong, J. Feng, J. Cao
Non-destructive measurement of residual stress distribution as a function of depth in sapphire/Ti6Al4V brazing joint via raman spectra
Ceram. Int., 45 (3) (2019), pp. 3284-3289
View PDF View article View in Scopus ↗ Google Scholar ↗

[45] Y. Ma, X. Bao, Z. Sui, X. Zhao, X. Liu
Quantifying Mg-Al cation distribution in $MgAl_2O_4$-spinel using raman spectroscopy: an experimental calibration
Solid Earth Sciences, 7 (1) (2022), pp. 60-71
View PDF View article View in Scopus ↗ Google Scholar ↗

[46] N.V. Minh, I.-S. Yang
A raman study of cation-disorder transition temperature of natural $MgAl_2O_4$ spinel
Vib. Spectrosc., 35 (1–2) (2004), pp. 93-96
View PDF View article View in Scopus ↗ Google Scholar ↗

[47] D. Hélie, M. Bégin, F. Lacroix, R. Vallée
Reinforced direct bonding of optical materials by femtosecond laser welding
Appl. Opt., 51 (12) (2012), pp. 2098-2106
View in Scopus ↗ Google Scholar ↗

[48] F.M.C. Capodacqua, A. Volpe, C. Gaudiuso, A. Ancona
Bonding of PMMA to silicon by femtosecond laser pulses
Sci. Rep., 13 (1) (2023)
Google Scholar ↗

[49] J. Fan, L. Yang
Damage mechanisms of bulk metallic glasses under high-rate compression
Int. J. Impact Eng., 106 (2017), pp. 217-222
View PDF View article View in Scopus ↗ Google Scholar ↗

Cited by (2)

Mechanisms and strategies for residual stress inhibition in laser-polished fused silica

2026, International Journal of Mechanical Sciences

Show abstract

Microstructure and mechanical properties of WC-Co/TC4 joints by square flat-top laser brazing

2026, International Journal of Refractory Metals and Hard Materials

Show abstract

View Abstract